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Whose Distance Matters? Own-Location versus Neighborhood Network Effects in a Planned City[2]

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Abstract:

This study examines how location shapes housing prices in Astana, the rapidly growing and largely planned capital of Kazakhstan. Using 118,660 residential sale listings collected in the first half of 2026, we estimate a point-level spatial hedonic model. We test two hypotheses: that price per square metre declines with distance from the central business district (CBD), and that the betweenness centrality of the nearest road segment is associated with price independently of CBD distance. Local Indicators of Spatial Association with false discovery rate correction show that 40.7% of listings belong to statistically significant price clusters. Lagrange Multiplier diagnostics confirm strong spatial dependence in OLS residuals. Among the SAR, SEM and Spatial Durbin (SDM) specifications, the SDM performs best: it removes residual spatial autocorrelation and reduces prediction error by about one third relative to OLS. The decomposition of SDM effects shows that prices fall by roughly 5% per kilometre from the CBD, and that this gradient operates mainly through neighbourhood spillovers rather than a dwelling's own location. Proximity to high-betweenness roads carries a small, strictly local price penalty, consistent with traffic disamenity rather than an accessibility premium. The findings extend tests of the monocentric model to an underexplored post-Soviet context. They also suggest that improving connectivity at the neighbourhood level and mitigating noise along major corridors are relevant levers for urban planning in Astana.

Keywords:

hedonic pricing; spatial Durbin model; monocentric city; betweenness centrality; spatial spillovers; Astana

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Introduction

After the collapse of the Soviet Union, different processes emerged in urban structure and city planning, where the shape of the city shifted from planned industrial structure to organic market oriented city with . Urban economists suggest that location within the city is a crucial determinant of housing prices. While classical literature suggests that distance to the central business district is negatively associated with house price(Alonso, 1964;Mills, 1972; Muth, 1969).

The model of monocentric city is tested in many North American, European, and developing countries, but post-Soviet and Central Asian countries' cities remain absent in the literature. Astana, the capital of Kazakhstan is informative case: largely planned, rapidly expanded in the last three decade, with well established commercial and administrative city center

In this study we attempt to address the relationship of dwelling price with distance to CBD and its location within the road network. To analyse this we used data of residential listing sales that were created in the first half of 2026.

We documented price clustering with Local Indicators of Spatial Association to determine any spatial clustering. We estimate OLS, SAR, SEM, SDM and interpret the preferred Spatial Durbin Model through its direct, indirect and total effects.

Results support our hypothesis, but with some nuances. Prices declined approximately to 5% with distance to CBD increasing by 1 kilometer, but this gradient operated through the neighbourhood's position rather than the housing unit itself. The paper thus provides one of the first large-sample spatial hedonic analyses of a Central Asian capital and shows that effect decomposition reveals location mechanisms that standard hedonic models miss.

The paper proceeds as follows. Section 2 reviews the literature, Section 3 describes the data, Section 4 sets out the methodology, Section 5 presents the results, Section 6 discusses them, and Section 7 concludes.

Literature Review

Hedonic price model and spatial dependence

The housing market can be one of obvious examples of hedonic price modelling. The theoretical foundation provided in work (Rosen, 1974) suggests that the price of goods is determined by characteristics of implicit attributes. Lancaster (1966) argues that demand for given goods is based on product characteristics rather than the product itself, this theory comes from ideas that goods themselves do not give utility to consumers, they possess characteristics and these characteristics give rise to utility (Strotz, 1957; Gorman, 1959). Based on that, housing is perceived as a bundle containing non-separable characteristics, treated as a heterogenic product with different characteristics with implicit utilities. Therefore, hedonic prices estimated through implicit

prices of properties of goods, can be applied to the real estate market. In Rosen's (1974) work, econometric models can be used to estimate hedonic price function, the sensitivity of price to its characteristics and then analyze demand on given properties.

The credit of pioneers who applied hedonic price models in residential property gives to Ridker and Henning (1967), they investigated association between air quality and property values. Freeman (1979) who noted and theoretically justified the appropriation of usage hedonic price modelling for the housing market. Typically the characteristics of the housing market are divided into three categories: locational, structural, and neighbourhood(Chau & Chin, 2003).

Different literature reviews are provided comprehensive reviews(Chau & Chin, 2003; Malpezzi S, 2003). The most usage cases of hedonic price modelling are linear or log, and can be estimated through econometric regressions. Even Lancaster's(1966) theory assumes a linear relationship between the price of goods and the characteristics contained in those goods, 'the log form model is often used as it has better goodness-of-fit and the advantage of correcting for heteroscedastic tendencies'(Wen & Tao, 2015).

Structural properties related to housing attributes. The single most important factor is floor area. Age and floor level are also important factors in apartments. Even the age and floor size relatively have stable effects on price level, the floor and other structural attributes significance can differ by city or over nation.

If we consider locational and neighbourhood characteristics of housing unit, there is a potential problem, arising from spatial effects. Spatial effects are widely discussed in literature(Anselin, 2003). Due to the spatial nature of housing units, as Tobler(1979) said: "Everything is related to everything else, but near things are more related than distant things", when houses are relatively close, the other units are correlated. The standard OLS is not appropriate in this scenario due to dependency of variables to each other and heteroskedasticity. "The consequences of spatial autocorrelation are the same as those of time series autocorrelation: the OLS estimators are unbiased but inefficient, and the estimates of the variance of the estimators are biased " (Dubin, 1998). The spatial dependence in the housing market is confirmed in many different ways, especially persistence of autocorrelation in various urban contexts(Basu & Thibodeau, 1998) . In order to work with spatial effects, there are many tools and models developed.(Anselin & Lozano-Gracia, 2009; Anselin, 2010).

Spatial models are performed better empirically compared to hedonic price modelling(Osland, 2010).The following subsections examine local spatial association indicators and the specifications of the spatial econometric models used in this study, as well as discuss transport accessibility as a location-determining factor.

Center of city and house prices

In classical urban economics, location within an urban area is a crucial determinant of housing price, which reveals that the spatial structure of a city has a significant impact on residential price. The model of monocentric distance, trade off

between distance cost and rent price established by these works: Alonso (1964), Mills (1972) and Muth (1969). They show a negative relationship between distance to the city center and housing price, these price disparities should be compensated with transportation cost.

This monocentric model is widely used in different urban structures, countries and economies. The center of the urban area is mostly the business district with highest density, and it shows a gradient decent structure of population, infrastructure and density. The persistence of literature shows the existence of this structure, for example in the US(McDonald, 1989), Japan(Zheng, 1991), Israel(Stern, 1993), these articles shows validity of monocentric model in different economies. The monocentric hypothesis is applied to different economies, and recently shown validity of standard urban model in USA(Paulsen, 2012), Oueslati et al.(2015) shown in 282 European cities, Jedwab et al.(2021) for 1010 cities in developing countries.

However in contrast to monocentric theory, another theory is developed. If monocentric theory assumes it is one center of an urban area, polycentric theory states that there are many “subcenters”(Fujita & Ogawa, 1982). Existing literature empirically shows recent shifts of many urban areas to polycentric models(Louf & Barthelemy, 2013) in different cases(Huang et al, 2015; Muñiz et al, 2008). Howerer, even empirical researches show changing dynamics in urban planning structure, in cities especially without conditions to emergence of subcenters, like population size and commuting cost(McMillen & Smith, 2003), the monocentric theory shows stable results.

While performance of monocentric urban structure widely tested across the different economies(Liotta et al, 2022), research on post-soviet and central asian contexts are still limited. Such types of cities are centralised city planned urban development and newly put in appearance central business districts are still underrepresented in literature. Astana, the new capital city of Kazakhstan, has rapidly expanded in the last three decades, offering opportunities for tests of compact monocentric urban structure, with well defined cultural, commercial city centres, without historically established subcenters.

Local Indicators of Spatial Association (LISA)

Even though monocentric models explain the systematic gradient decline in prices from the CBD, localized clusters or subcenters in urban morphology can still emerge. Some amenities of a city can positively or negatively affect neighborhood prices. As previously represented as a problem of hedonic price model, spatial data requires further investigation(Dubin, 1998). To explore this clusterization, authors use global Moran’s I tests, to determine any autocorrelation between objects, as well in real estate research(Emmer et al, 2026). While classical Moran’s I test shows the presence of spatial structure, to spot clusters mostly used Local Indicators of Spatial Association or LISA. Method developed by Anselin(1995), which gives a LISA cluster map, containing four established categories.

Applying LISA methodology to the real estate market is one of the common parts of explanatory analysis(Dejniak,2018;Barreca et al, 2018). Still because LISA is computed simultaneously across a large number of spatial units, applying a conventional significance benchmark ($p < 0.05$) to each unit independently inflates the expected number of false positives. While the Bonferroni correction addresses this by controlling the family-wise error rate, it does so conservatively, substantially reducing statistical power in applications with hundreds of simultaneous tests. Following common practice in LISA-based spatial cluster detection, this study instead applies the false discovery rate (FDR) correction (Benjamini & Hochberg, 1995), which controls the expected proportion of false discoveries among identified clusters while retaining greater power to detect genuine spatial patterns.

LISA analysis is part of the explanatory analysis, it shows clusters but does not give proper explanation and mechanisms behind that. To analyze spatial nature of urban morphology, literature used different spatial econometrics methods, like spatial error model(SEM),spatial durbin model(SDM), Spatial Autoregressive(SAR)

Spatial Econometrics

Spatial econometric is multiple methods which can be used to analyse spatial nature of data. While LISA analysis shows explanatory clusters, spatial econometrics takes into account these clusterization and dependency of spatial structures (Anselin, 2022), and allows to estimate effects of predictors which are related to each other.

Common regression models like OLS models are not appropriate, as Dubin(1998) pointed earlier. After Moran's I test, Lagrange multiplier test provide the possibility to test the presence of spatial errors or autocorrelation(Anselin, 1988).

The variants of the spatial model that are applicable include spatial lag model (SAR) and the spatial error model (SEM). Spatial correlation among the dependent variables is defined as a spatial lag situation which is specified by the spatial lag model. When spatial dependence exists in the error term, a spatial error model is employed. Further generalizations of two models called Spatial Durbin Model (SDM), which incorporate both spatial lags of the dependent variable and spatial lags of the explanatory variables in addition to the conventional explanatory variables(LeSage and Pace, 2009)

The selection of certain model between SDM, SAR, SER usually justified with Lagrange Multiplier diagnosis test proposed by Anselin(1988), applied to the residuals of an initial OLS regression. To determine which type of spatial dependence dominates, adjusted versions statistic can be used, due to other dependencies can inflate LM test for a spatial lag and a spatial error. If both the lag and error will be statistically significant after adjustments, Spatial durbin model is preferably used to avoid strict restrictions of SAM and SEM. To further choice are supported with information criterias(AIC, BIC) and log-likelyhood.

Transport betweenness

While classical monocentric theory takes Euclidean distance from the central business district to the point as a preliminary determinant of the price, network science offers different methods to characterize location within street networks. One of them, closeness centrality offers a conventional method to conceptualise accessibility in terms of how effectively certain locations can be reached within the road network. Another method, betweenness centrality , does not measure how effective it can be, but how the given segment lies on the shortest path between any two points in the network. A housing located near segment with high betweenees centrality not necessarily more accessible, but it exposed traffic corridor which can be as advantage like ease of movement or commercial activity or drawbacks as noise, pollution.

Because of dual nature, the effect is ambiguous. Chakrabarti et al(2022) analyses 400 neighborhoods in Kolkata and shows that centrality measures are included in price level of houses. Using a closely related methodology, the space syntax literature computes betweenness ("choice") at the level of individual street segments and assigns this value to nearby buildings (Law, 2018).

Taken together, evidence shows that the betweenness is unlikely to show positive and universally effect, which is consistent with hedonic literature, where high traffic loads are negatively associated with the price of housing through noises and pollution.

Hypothesis:

H1: Euclidian distance to the CBD is negatively associated with house price.

H2: The betweenness centrality of the nearest road segment associated with housing price independently of distance to CBD. The direction of this association is not predicted in advance , due to ambiguity between accessibility or traffic disamenity.

Data

Study Area

Astana, the capital city of Kazakhstan from 1997, is an important economical, political, cultural center of the country. It is a nearly new city, which rapidly urbanized

and expanded in the last three decades, after the relocation of the capital city from Almaty. According to data from the Bureau of national statistics, the population increase from 2006 to 2016 was 108873 individuals. The residential buildings are also rapidly expanding, where only in the period from January 2026 to May 2026 the area of commissioning residential area was 1561.1 thousand square meters. In the date of August 2026, in Astana exist 6 districts: Esil district, Baikonur district, Almaty District , Saryarka District , Sarayshyk District which will be newly established in 2025.

We chose our CBD as coordinates of the monument of Baiterek, which corresponds to 51° 7′ 42″ N, 71° 25′ 50″ E, is a tourist attraction observation tower, and it is emblematic of the capital city. The name “Baiterek” means “Tree of Life.” The monument carries a deep philosophical meaning that reflects the worldview of the ancient nomads. Near Baiterek are located many shopping centers, administrative buildings and skyscraper offices.

Data Collection and Cleaning

We chose a single housing option as the basic analysis unit. These housing units can be either apartments in blocks, cottages, any area for living located within the urban area and mostly connected with roads. The housing sample data obtained from online listing service for real estate. We parse all listings created from 1st January 2026 to mid of June 2026. The number of raw listings is 248888. We parse information about the price in Kazakhstani tenge, flat area, the age of building, ceiling high, material of building, coordinates.

While services contain all types of residential property areas like office, housing, commercial etc. and types of bid(rent, sell) options under overlapping categories, we restricted our data rental listings by excluding filter keywords in listing title(“per month” and “per day” in Russian). Our listing also must required to report positive, valid price per square meter, valid geographic coordinates, coordinates must be within a bounding box of Astana city(51.00° - 51.25° N, 71.30° - 71.61° E); and a complete set of structural variables: floor, flat area, ceiling high, building material, building year. Finally, extreme values in price per square meter were trimmed at 2.5 and 97.5 percentiles due to prevent data entry errors and non-representive listings. The full step by step changing in number of observations are represented in Table 1:

Table 1

Step	N	% from base
All listings in data set	248888	100
Selling housing units	161062	64.7
With valid prices	161062	64.7
With valid coordinates	161062	64.7
Within geographic borders	159731	64.2
With all structural values	124900	50.2
After trimming (2.5:97.5	118660	47.7

percentile)		

Variable construction

Following cleaning the data, we constructed several variables for further analysis. Building age was computed by the difference between the present year(2026) and building year, then we mean centered to make it easier to interpret, and to reduce the collinearity with its squared term, which is included in our analysis to show a nonlinear relationship. Building material treated as categorical variable. We transformed the price per square meter and the living area was transformed into logarithmic form to reduce skeweness and allow us to interpret coefficients in percentages.

Many listings share identical same coordinates, because they correspond to the same residential building complex or apartments. This issue will construct problematic neighbours in further analysis, we applied Gaussian jitter to each point coordinates. In computed distance in kilometers from central business district, such displacement insignificant for substantive results in solving the numerical problem.

The road network was obtained from OpenStreetMap(OSM), only line geometries with tag ”highway” were taken into analysis, and the network was restricted only to road classes: motorway, trunk, primary, secondary, tertiary, residential and unclassified roads. Pedestrian and service ways were excluded due to they were not relevant to vehicular traffic flow.

The road segments were converted into an undirected spatial network, and pseudo-nodes - vertices of degree two where a road continues without an intersection were collapsed through spatial smoothing. This step prevents a single continuous street from being artificially fragmented into many short edges purely as an artifact of how OpenStreetMap digitizes roads, which would otherwise distort centrality estimates by inflating the apparent number of distinct segments along a single corridor. The number of connected components in the resulting network was checked. Betweenness centrality was computed at the edge level, using each edge's length as the shortest path weight, so the centrality proximity weighted shortest path usage in meters. Direction of travel is not modelled, because of an undirected network.

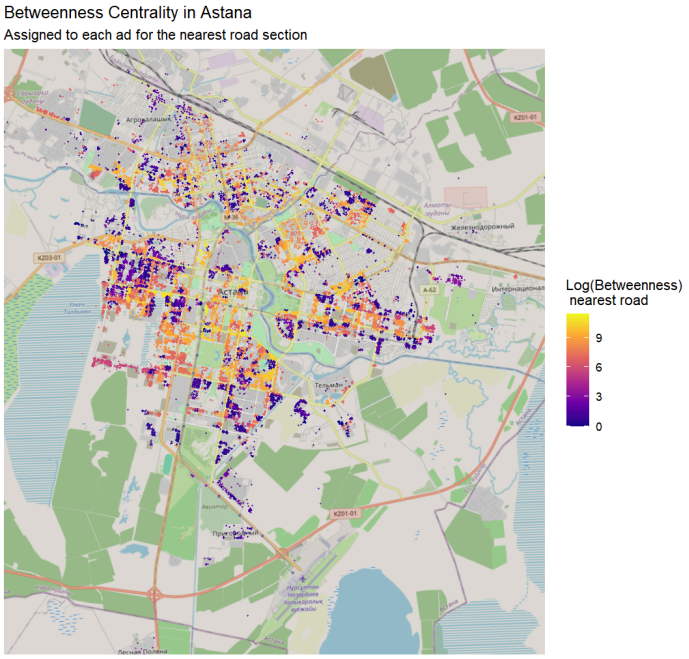
Each housing listing was then assigned with the betweenness value of nearest road segment, identified via nearest feature matching. Due to the nature of betweenness centrality, to prevent right-skewed distribution with most values equal to zero, the values were transformed in log(x+1) form. The distribution of betweenness centrality shown in Figure 1.

Methodology

Point level spatial regression analysis requires a defined neighbours to construct the well organised spatial weights matrix. (Getis & Aldstadt, 2004). In this study we

applied the k-nearest-neighbour(KNN) method graph to connect nearest eight(k = 8) observations, which in our case are listings, regardless of distance. Dense listing is one of the properties in highly urbanized spatial density structure, but KNN method guarantees constant, computationally computable number of observations. However such method has not a unobjectable property, as Kubara and Kopczewska (2024) show in their paper empirically and simulationally, that model fit, they used AIC, is a nonlinear function of k, and appropriate chosen k can reduce bias to approximately to 20%.

Figure 1



Before estimating the regression models, we examined spatial outcomes directly using Local indicators of Spatial Association(Anselin, 1995). Log price per square meter for each listing was chosen for computation of Local Moran's I, based on the KNN spatial weight matrix. For each unit of observation, the statistic compares its own standardized value to the weighted average of its neighbour values, allowing classifications into four classes: High-High, High-Low, Low-High and Low-Low. High-High class means a high-pricing listing surrounded by high-price neighbours, Low-Low vice versa, low-pricing listings surrounded by low-price neighbours. High-Low, Low-High are two spatial outliers.

Because this test is performed in a very large number of units, a nominal significance threshold of alpha = 0.05 would be expected to flag several thousand locations as significant by chance alone, and the underlying tests are not independent across neighboring locations by construction. The developers of spatially dependent test correction,Caldas de Castro and Singer (2006) show that p-values were adjusted using the false discovery rate procedure before classifying a location as belonging to a

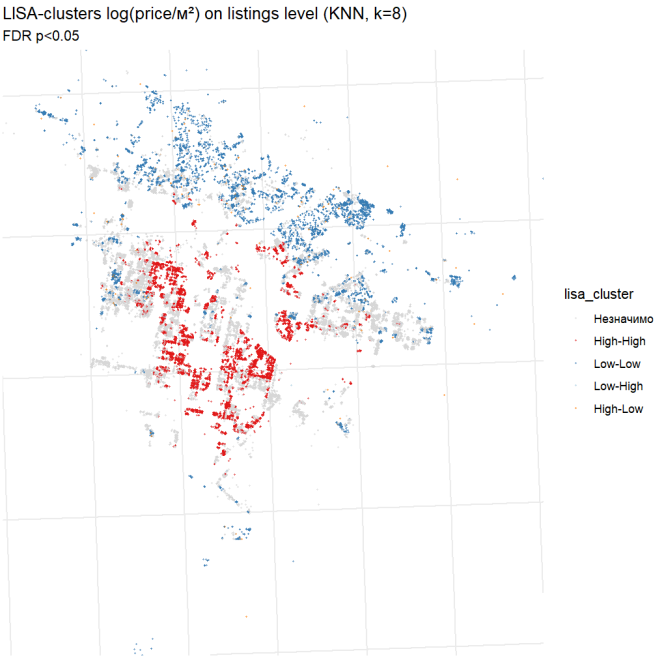
significant cluster, locations that did not remain significant after adjustment were classified as non-significant.

This analysis serves two purposes. Firstly, in a spatial econometric approach, the existence of spatial clustering should be predetermined, this is a necessary condition. Secondly, it gives descriptive visual evidence of clusterization in Astana city. The result shown in Figure 2.

Model specification

The baseline specification is hedonic equation where the dependent variable is log transformed price per square meter in KZT and explanatory variables includes structural characteristics of housing unit and location within the road network and distance to CBD:

Figure 2



Because logarithmic form NearestBetweenness and Distance to CBD both captures spatial aspects of a property's position within the city and major arterial network tend to concentrate toward the center, these two variables was checked via their pairwise correlation and via their variance inflation factor (VIF) computed on the baseline OLS fit. In our case as shown in Table 2, there is no meaningful multicollinearity between these two variables(1.03 and 1.10, respectively).

Table 2

Variable	GVIF	Df	GVIF^(1/(2·Df))
building_age_c	2.484	1	1.576

building_age_c²	2.657	1	1.630
log_area	1.142	1	1.069
floor	1.077	1	1.038
ceiling height	1.027	1	1.013
material	1.504	3	1.070
log_betweenness_nearest	1.057	1	1.028
dist_cbd_km	1.219	1	1.104

The baseline model is firstly estimated by OLS. Because the assumptions are not held to spatially distributed housing units, OLS residuals are tested for spatial dependence with Lagrange Multiplier tests(LM). The standard LM tests for a spatially lagged dependent variable (LMlag) and for spatially autocorrelated errors (LMerr), together with their robust counterparts (RLMlag, RLMerr), which remain valid when both forms of dependence are potentially present simultaneously. The tests were computed using a spatial weight matrix.

All four tests reject the null of no spatial dependence decisively Table 3:

Table 3

Test	***Statistic***	***df***	***p-value***
LM-lag (RSlag)	*156,300.5*	*1*	$< 2.2\times10^{-16}$
LM-error (RSerr)	*175,835.9*	*1*	$< 2.2\times10^{-16}$
Robust LM-lag (adjRSlag)	*1,522.9*	*1*	$< 2.2\times10^{-16}$
Robust LM-error (adjRSerr)	*21,058.2*	*1*	$< 2.2\times10^{-16}$

In this pattern, where both robust statistics remain significant rather than one dominating the other, does not point unambiguously to either a pure SAR or a pure SEM specification. It is nonetheless notable that the robust LM-error statistic is roughly an order of magnitude larger than its lag counterpart, suggestive of error dependence as the stronger of the two forms, but given that both remain significant, this diagnostic evidence alone is not treated as conclusive. Instead, this ambiguity motivates estimating all three specifications : SAR, SEM, and SDM.

Model selection

Given the ambiguous pattern after the diagnosis, where both standard and robust LM tests for lag and error dependence were jointly significant, we cannot rely on the diagnostic tests alone. We will follow a strategy supported by Elhorst (2010), where rather than a priori choosing a model between SAR and SEM based on LM diagnosis, the most general SDM is estimated, and after that on the basis of formal comparison the best model is determined.

Three models are estimated and compared. The SAR model allows the outcome itself depends on a spatially weighted average of neighbour outcomes:

$$\mathrm{y} = \rho\,\mathrm{W}\mathrm{y} + X\beta + \epsilon$$

Where is spatial autoregressive coefficient, and W is the standardised KNN weights matrix. In context of application in housing price, Kim et al(2003) stated that prices of housing in listings are not determined independently, but the price flags transmit to other listings. For example, consumers' expectations affecting one listing's price are expected to partially transmit to its neighbour's prices as well.

The SEM model instead confines spatial dependence into to the error, leaving a direct relationship between X and y without being mediated by neighboring results:

$$\mathrm{y} = X\beta + \mathrm{u},\ \mathrm{u} = \lambda W\mathrm{u} + \epsilon$$

A significant implies that, unobserved, spatially correlated factors are omitted, neighbourhood's amenities, unmeasured infrastructure or micro-level, without implying the neighbour houses' price is not casually impacted to each other.

The SDM includes SAR and spatially lagged explanatory variables as regressors:

$$\mathrm{y} = \rho\,\mathrm{W}\mathrm{y} + X\beta + WX\theta + \epsilon$$

When , we will have a SAR model. The SDM model is the most general among three specifications, which cover both direct and indirect characteristics of the housing unit, conditions that are relevant for our study's location variables.

All three models are estimated by maximum likelihood using the `spatialreg` package in R language. Estimating this likelihood requires computing the log-determinant of (I − ρW), a step whose feasible computational strategy depends on both the size and the structure of W (Bivand, Hauke & Kossowski, 2013). The default eigenvalue-based method is computationally infeasible at N ≈ 118,660. A sparse Cholesky decomposition is a standard alternative for large N, but requires a symmetric weights matrix; because the KNN weights matrix is row-standardized, and a KNN neighbor relation is not itself guaranteed to be symmetric (point A having point B among its k nearest neighbors does not imply the reverse), W is not symmetric here. Estimation therefore used an LU decomposition, which does not require symmetry and remains computationally tractable for the large, sparse weights matrix used in this study. The

choice among them was made on Akaike and Bayesian information criteria, likelihood-ratio.

In SAR and SDM , the coefficients can not be interpreted as marginal effect of that covariate price, because the presence of spatial neighbour dependency, which creates a feedback loop: changing in one's listing characteristics to the price will lead to change neighbours' prices through spatial lag, which in turns affects to the listings price itself, and so on to whole network. Following LeSage and Pace (2009), this effect can be divided into direct and indirect effects, where sum of them is total effect. This decomposition is applied to SDM, since in this model it is actually used for substantive interpretation.

Computing the exact effects requires a substantial amount of computational resources, which require materializing matrices on the order of N^2. Traces were therefore approximated using the Monte Carlo method of Barry and Pace (1999). Standard errors and significance for the direct, indirect and total effect obtained via simulation. The resulting direct, indirect, and total effects, together with their simulated significance levels, are reported and interpreted in the Results section.

Results

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Table 4 reports the baseline OLS estimates of the combined hedonic equation. The model explains a 46% of variation in log price per square meter. Results are consistent with a specification that omits any location fixed effects(beyond two that we specify in our OLS model). All structural values are significant at conventional level, except the material types(1 and 2). Building age shows the expected nonlinear relationship, where implied quadratic term specification; turning point approximately 30.5 years above the average sample mean age, plausibly reflecting heritage premium for the oldest buildings.

Variable that indicates log area of the housing unit carries negative sign coefficient. Since in our analysis the response variable is log price per square meter rather than total price, this is consistent with "size discount" in the housing market, whereby larger units command a lower unit price even as total price rises with size. Results for ceiling height and floor have expected directions.

Both locational predictors are significant and have expected signs. Distance to CBD has a negative association, consistent with H1 and the classical monocentric prediction that prices decline with distance from the CBD. Variable responsible for proximity to the high-betweenness road segment is associated with lower prices, which are consistent with traffic-diamenity side of the H2, rather than an accessibility premium.

However, OLS estimates are not taken into account for spatial dependency among observations. Further sections will show that this dependency is substantial. The OLS estimates in this section should be read as a benchmark, motivating the spatial modelling, rather than as the final results. Regarding multicollinearity, VIF values remains below (Table 2), so none of OLS estimates reported here are materially distorted by collinearity among the regressors.

Table 3 shows results of Lagrange Multiplier tests, which will contribute as evidence of spatial dependence in OLS residuals. Standard and robust versions of LM-lag and LM-error are extremely large and critical. This pattern rules out the OLS estimates in Table 4, OLS standard errors are unreliable, and the point estimates themselves may be biased.

Table 4 reports AIC for the three spatial specifications. The SDM achieves the lowest AIC, despite its additional degrees of freedom, indicating that improvements in fit from including spatially lagged variables exceed the effect of added variables complexity. Moran's I on the OLS is strongly positive and highly significant, confirming the extensive spatial dependency already indicated by LM tests. After estimating the SDM, residual Moran's I falls to -0,041 and is no more significant. OLS left unexplained in its residuals has been almost entirely absorbed by the SDM model. Taken together, the AIC comparison, the LR test, and the residual diagnostics converge on the same conclusion: the Spatial Durbin Model is the preferred specification, and all subsequent interpretation is based on it.

Because the SDM was selected as the preferred model, its raw coefficients cannot be read directly as marginal terms. Table 5 reports the direct, indirect and total effects obtained via Monte Carlo decomposition.

Distance to CBD shows a striking divergence between its direct and indirect components. The direct effect is positive and significant, however the indirect effect is large, negative and highly significant . This means listing further from CBD itself is associated with higher price, but spillover effect drives price to the opposite direction. Substantively, a genuinely spatial, network mediated form of the CBD distance gradient rather than a purely individual location effect.

Log form of betweenness of nearest road shows the opposite pattern: with the direct effect being negative and highly significant, the indirect effect is close to zero and not statistically significant. The total effect is negative and significant at the 5% level, driven essentially entirely by the direct component. Taken together, this pattern supports H2's traffic disamenity interpretation, where road-network betweenness is strictly local, as a direct exposure cost like noise and traffic.

Table 4

	OLS	SAR	SEM	SDM
Constant	13.658***	3.347***	13.903***	3.016***
	(0.007)	(0.028)	(0.008)	(0.028)

	OLS	SAR	SEM	SDM
Age	-0.022***	-0.005***	-0.012***	-0.008***
	(0.000)	(0.000)	(0.000)	(0.000)
Age²	0.000***	0.000***	0.000***	0.000***
	(0.000)	(0.000)	(0.000)	(0.000)
log(Flat area)	-0.039***	-0.051***	-0.088***	-0.088***
	(0.001)	(0.001)	(0.001)	(0.001)
Floor	-0.003***	-0.002***	-0.001***	-0.001***
	(0.000)	(0.000)	(0.000)	(0.000)
Ceiling Height	0.018***	0.007***	0.003***	0.004***
	(0.001)	(0.001)	(0.001)	(0.000)
Material 1	-0.001	0.007***	0.000	0.001***
	(0.003)	(0.002)	(0.003)	(0.000)
Material 2	0.005	0.006*	0.009**	0.009***
	(0.005)	(0.003)	(0.004)	(0.002)
Material 3	0.020***	0.020***	0.019***	0.020***
	(0.003)	(0.002)	(0.002)	(0.001)
log(Betweenness nearest road	-0.001***	-0.000*	-0.001***	-0.001
	(0.000)	(0.000)	(0.000)	
Distance to CBD, км	-0.059***	-0.016***	-0.058***	0.023
	(0.000)	(0.000)	(0.001)	
W × Age				0.003***
				(0.000)
W × Age²				-0.000
W × log(Area)				0.094***
				(0.002)
W × Floor				-0.000
W × Ceiling Height				0.009***
				(0.000)
W × Material 1				-0.005
W × Material 2				-0.009
W × Material 3				-0.019***
				(0.001)
W × log(Betweenness)				0.001***
				(0.000)
W × Distance to CBD				-0.036
ρ		0.767***		0.773***

	OLS	SAR	SEM	SDM
		(0.002)		(0.002)
λ			0.802***	
			(0.002)	
Num.Obs.	118660	118660	118660	118660
R2	0.460			
R2 Adj.	0.460			
AIC	-76894.4	-152338.9	-154112.4	-156300.5
BIC	-76778.2	-152213.0	-153986.5	-156077.8
Log.Lik.	38459.191	76182.469	77069.204	78173.264
F	10122.354			
RMSE	0.17	0.12	0.12	0.12
• p < 0.1, ** p < 0.05, *** p < 0.01				

Table 5

Variable	***Direct***	***Direct SE***	***Indirect***	***Indirect SE***	***Total***	***Total SE***
Building age (centred)	*−0.0090****	*(0.0002)*	*−0.0138****	*(0.0002)*	*−0.0228****	*(0.0003)*
Building age² (centred)	*0.0001****	*(0.0000)*	*0.0003****	*(0.0000)*	*0.0004****	*(0.0000)*
log(Floor area)	*−0.0810****	*(0.0012)*	*0.1086****	*(0.0053)*	*0.0275****	*(0.0058)*
Floor	*−0.0018****	*(0.0001)*	*−0.0054****	*(0.0003)*	*−0.0072****	*(0.0004)*

Ceiling height	*0.0072***	*(0.0005)*	*0.0513***	*(0.0040)*	*0.0585***	*(0.0043)*
Material (cat. 1)	*0.0003*	*(0.0003)*	*−0.0162****	*(0.0022)*	*−0.0160****	*(0.0022)*
Material (cat. 2)	*0.0085***	*(0.0030)*	*−0.0093*	*(0.0081)*	*−0.0009*	*(0.0108)*
Material (cat. 3)	*0.0189***	*(0.0009)*	*−0.0138****	*(0.0042)*	*0.0051*	*(0.0045)*
log(Betweenness of nearest road)	*−0.0010****	*(0.0002)*	*−0.0000*	*(0.0004)*	*−0.0011**	*(0.0006)*
Distance to CBD, km	*0.0188***	*(0.0004)*	*−0.0732****	*(0.0009)*	*−0.0544****	*(0.0010)*

Table 6 reports the distribution of listings across LISA cluster type for log price per square meter. Of the 118,660 listings, 48,321 (40.7%) fall into a statistically significant spatial cluster after FDR correction. Among significant clusters, the overwhelming majority are positive-autocorrelation types: 25,076 listings (21.1%) form Low-Low clusters and 22,566 (19.0%) form High-High clusters, together accounting for 40.1 percentage points of the sample. The distribution is consistent with the strong global spatial dependence, previously documented, prices are not distributed randomly across city, but form clusters with consistently high or low prices.

Table 6

Cluster	N	%
Незначимо	69,864	58.9
High-High	22,566	19.0
Low-Low	25,076	21.1
Low-High	469	0.4
High-Low	685	0.6

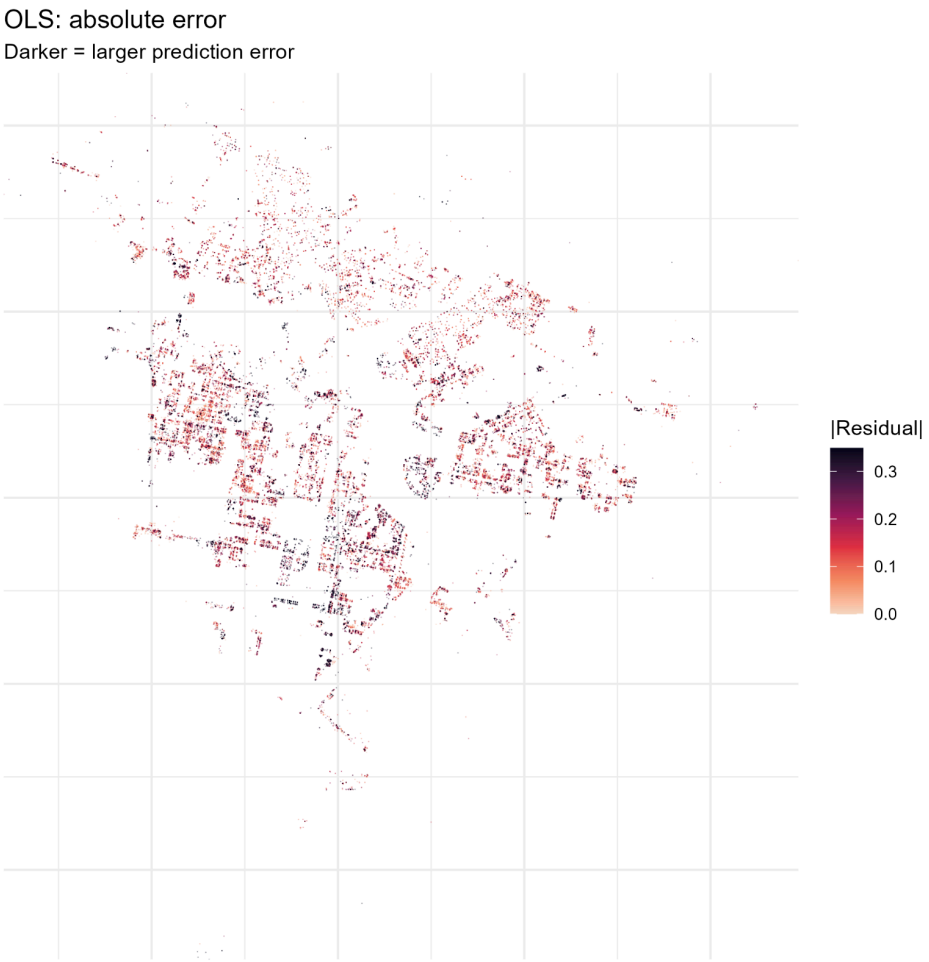
Cluster	N	%
Total	118,660	100.0

Table 7 reports overall predication accuracy for OLS and SDM, where SDM achieves substantially lower values of RMSE and MAE on log price per square meter. This improvement consistent with comparison results in previous sections that confirms accounting for spatial dependence yields substantively, typical prediction error shrinks a third once spatial structure is modeled explicitly. The signed residual maps (Figure 3, Figure 4) show where each model is over or under predicting price, and illustrate how including spatial factors change residuals. The actual-vs-predicted scatter (Figure 5) for the SDM provides a final visual check of overall fit: points clustering tightly around the 45° line indicate accurate prediction, while systematic deviation from it at the extremes (very high or very low prices) would indicate the model's remaining limitations, particularly for atypical listings not well described by the included characteristics.

Table 7

Model	RMSE	MAE
OLS	0.1750	0.1372
SDM	0.1196	0.0904
Improvement	-31.6%	-34.1%

Figure 3



Discussion

This study tested two location based hypotheses about housing prices in Astana using a point-level spatial hedonic model estimated on 118 660 listings. H1 - that distance to CBD is negatively associated with price, is supported in its total effect. However SDM decomposition reveals that this masks a more complex mechanism: the negative gradient is carried almost by indirect network spillover effect, in fact direct effect is positive. H2 - that road network betweenness centrality is associated with price in ambiguous sign is also supported, the direct effect is negative, but indirect positive that is not statistically indistinguishable from zero.

The CBD distance gradient as phenomena

The classical monocentric city model (Alonso, 1964; Muth, 1969) suggests a negative own location gradient, a property's price should decline with its own distance to the central business district, and this decline should be applied only for that property alone. The positive direct effect found here complicates this picture. It implies that, holding the surrounding neighbours' network fixed, the distance to CBD is not a penalty. What drives the aggregate negative relationship is instead the composition of the surrounding network: living in a neighborhood that is, on average, far from the center depresses price far more than one's own individual distance does.

This finding resonates with broader literature on topic of monocentricity in rapidly transforming cities. Wen and Tao (2015), studying Hangzhou during a period of intense urban transformation, similar in speed with Astana's post 1997 growth, find that as the city's structure becomes more polycentric, the simple distance to CBD gives complex spatial patterns shaped by multiple centers of activity and their surrounding environment. Astana's own history as a purpose built, administrative and commercial center is still consolidating, and rapid growth of non-organic high-density developments produces a similar decoupling of "my own distance to the center" from "my neighborhood's overall position, which is consistent with our results. This suggests that the network decomposition adopted in this study may be particularly well suited for rapidly urbanizing cities with inorganic growth in general, where classical single-point distance measurements are least likely to fully reflect the true location position in the market.

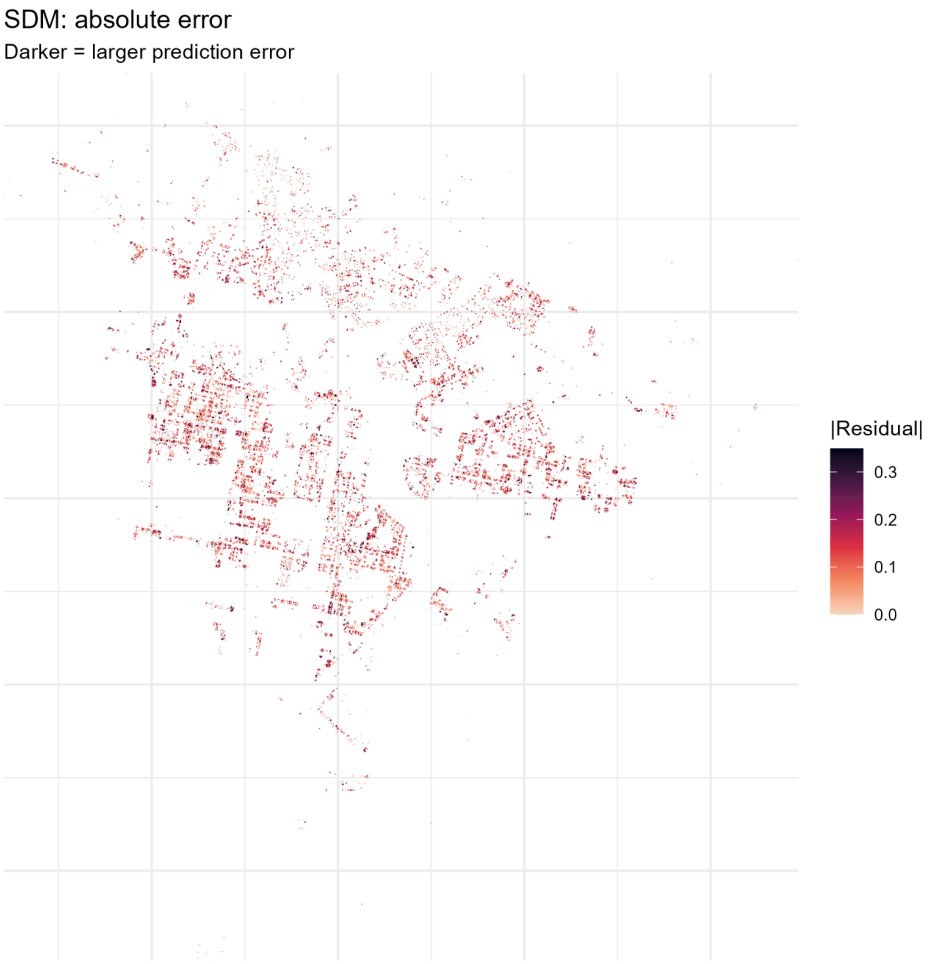
Betweenness Centrality as a Localized Disamenity

The betweenness results directly repeats the concept distinction that we state in literature review between betweenness and accessibility proper. Results are consistent with a disamenity mechanism like noise, pollution and inconsistent with an accessibility-premium mechanism. This join in with Anselin and Lozano-Gracia's (2009) broader finding that traffic-related disamenities are capitalized negatively into nearby housing prices. The study is also consistent with literature which states network-position effects are non-linear or spatially heterogeneous. This study's finding of a negative, spatially confined betweenness effect is one specific realization of that general non-uniformity, for one city and one specific network measure.

Practical applications

Two practical implications follow for urban planning and valuation practice in Astana. First, because distance to CBD operates through a neighborhood's average network position rather than single unit's own coordinates, area-wide interventions that improve a neighborhood's effective connectivity to the center are likely to have a broader capitalization effect. Second, negative confined betweenness association implies localised price penalty for proximity to major roads traffic corridors, is not offset by any detectable accessibility benefit, repeating mitigation-effect logic documented in the traffic-noise hedonic literature (Swoboda et al., 2015)

Figure 4



Conclusion

This study analysed the spatial structure of housing prices in Astana using point-level hedonic model estimated on listings from 2026. The results confirm that the city's housing market is spatially structured, after false discovery rate correction, ~40% of listings are statistically significant. Lagrange Multipliers tests justify usage of spatial models. The Spatial durbin model was the preferred specification on information criteria, it removed the residual spatial autocorrelation left by OLS and reduced prediction error (RMSE) by about a third.

Both hypotheses are supported, with important nuances. For H1, the total effect of distance is negative but SDM composition shows the gradient works through indirect components. For H2, betweenness centrality of the nearest road have no detectable spillover. This fits localised traffic disamenity, rather than accessibility.

These findings have practical implications for urban planning and property valuation in Astana city. Because center distance premium is associated with neighbourhood level, area wide connectivity interventions can be more effective rather than individually targeted. Methodologically, the study shows that ignoring spatial dependence leads to materially different and less reliable conclusions. Spatial econometric models with effect decomposition are especially well suited to rapidly

growing, planned cities, where single-point distance measures may not fully capture a dwelling's position in the market.

Future research could extend this analysis with using data from different cities, panel data to track how the gradient changes as Astana grows, and add additional amenities and accessibility measurements as public transportation, green zones, and educational organizations. Comparisons with other fast growing cities would show whether the patterns found here are specific to Astana or general to central planned urban development.

Figure 5



Limitations

Several limitations should be considered. First, the analysis relies on asking prices from online listings, which can differ from actual prices. Second, a single cross-section can not trace how gradient changes as Astana continues to expand. Third, hedonic specification omits several potentially neighbourhood attributes like public transport, Fourth, some results depend on modelling choices. The decomposition of the CBD distance effect into direct and indirect components should be interpreted

with caution. Finally, the estimated relationships are associations rather than causal effects, since road network structure and residential development are jointly determined.

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